



Zenith Boom

Ultralight Bistable Composite Structure

The Zenith Bistable is our flagship high-performance deployable structure, capable of holding two stable states (coiled and extended) without complex locks.

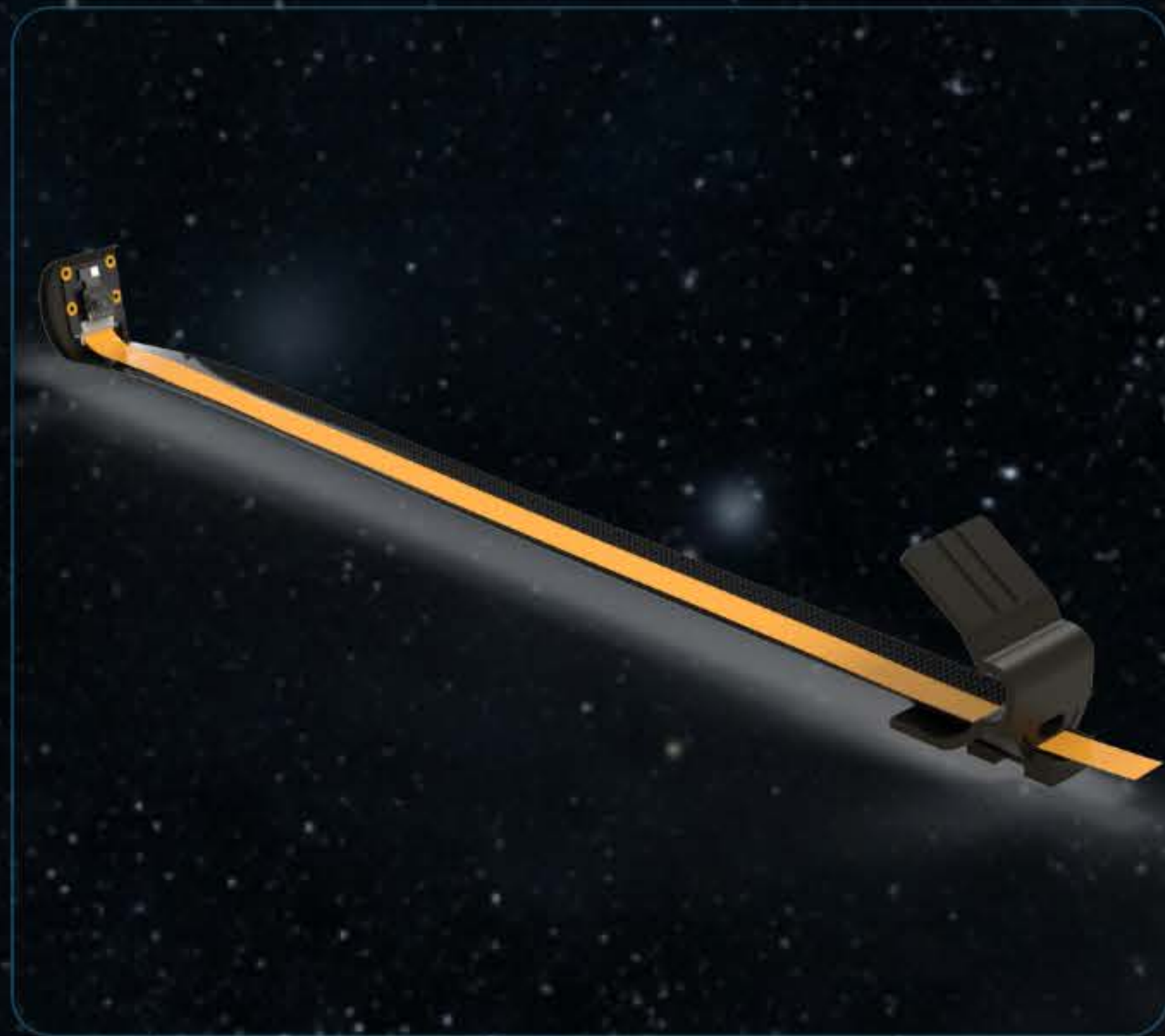
Engineered with BhorForce® advanced composites, it delivers exceptional reliability. The standard 50cm variant weighs just ~10g—lighter than a coin—yet supports a 120g payload, making it ideal for mass-constrained missions.

SPECIFICATIONS

Material	UD Carbon Fiber
Mechanism	Bistable / Strain Energy
Stowed Form	Coiled
Mass (50cm)	~10g
Payload	120g (1200% Mass)
Cross-Section	20-50mm (Custom)
Op. Temp	-40°C to +80°C
TRL	4 (Lab Component)

KEY FEATURES

Technology	Bistable Composite
Strength	High Stiffness-to-Weight
Applications	Antennas, Sensors
Variants	Bistable / Deployment



Scout Boom

Deployable Camera Mechanism

The Scout Boom is a deployable camera mechanism designed for capturing visual inspection data and "selfies" of the satellite in orbit.

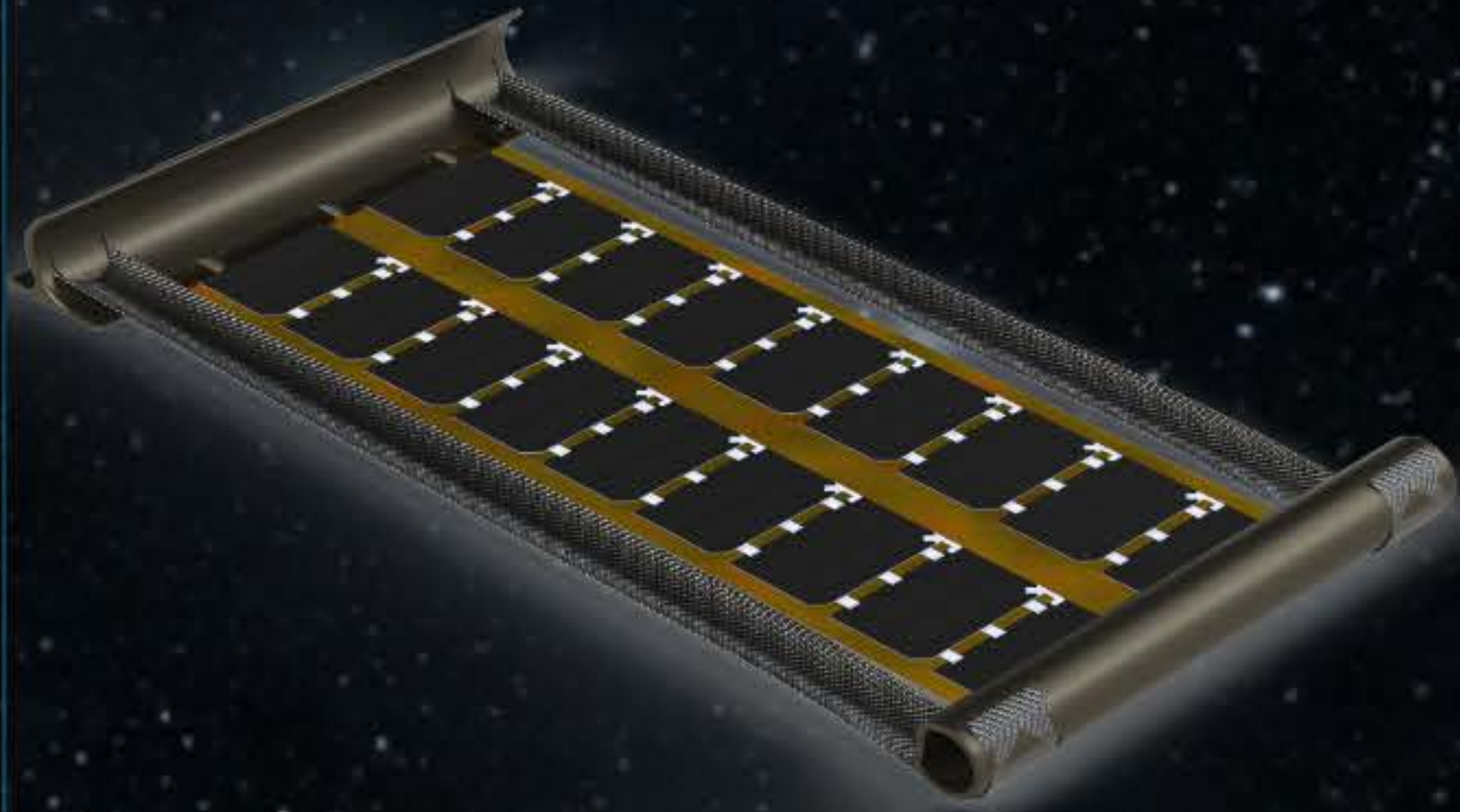
It features an integrated flex PCB running along the boom length, eliminating complex wire harnesses and reducing snag risks. The system uses stored strain energy to deploy a bistable composite boom rolled compactly against the CubeSat.

SPECIFICATIONS

Deployment	Stored Strain Energy
Form Factor	1U+ Compatible
Length	Up to 1.5 m
System Mass	< 150g (w/ Cam)
Tip Mass	40g (at 70cm)
Camera	5 MP (Typical)
Interface	MIPI CSI-2 / USB
Power	3.3V / 150mW

KEY FEATURES

Integration	Embedded Flex PCB
Actuation	Burn Wire (HDRM)
Feedback	Limit Switch
TRL	4 (Prototype)



Horizon Array

Next-Gen Roll-Out Solar Array

The Horizon Array is a deployable composite boom system inspired by the ISS iROSA. Its rolling mandrel design saves critical volume while acting as a protective core for the fragile solar blanket during launch.

By eliminating slip rings and enabling multi-functional utility, this system delivers high power density with simplified electrical architecture.

SPECIFICATIONS

Prototype Length	65 cm
Architecture	Rolling Mandrel
Power Transfer	Direct (No Slip Ring)
System Mass	< 250g (Est.)
Stowed Dims	300 x 50 x 50 mm
Deployment	Self-Deployable
Payload	~200g (Blanket)
Output	~80W

KEY FEATURES

Mechanism	Bistable Composite
Actuation	Burn Wire / Resistor
Feedback	Limit Switch
Variants	ROSA / FOSA



Aurora Sail

Tri-Boom Solar Sail Mechanism

The Aurora Sail is a precision-engineered deployment mechanism for high-area CubeSat solar sails. It utilizes a refined modular Tri-Boom architecture to maximize packing efficiency.

By integrating precision carbon fiber stabilizers, the system provides a low-friction, high-reliability solution for deploying large-scale membranes in complex orbital environments.

SPECIFICATIONS

Configuration	Tri-Boom
Boom Material	Composite Slit-Tube
Stowed Volume	~80mm Dia (Tuna Can)
Sail Area	0.425 m ² (Scalable)
Sail Material	CP1 Polyimide
Mass (System)	140g (excl. motor)
Actuation	NEMA 17 / BLDC
TRL	4 (Validation)

KEY FEATURES

Architecture	Modular Tri-Boom
Guide Rails	3mm Carbon Fiber
Actuation	Burn Wire Release
Feedback	Limit Switch